Hierarchical Machine Learning Modelling of Multi-phase Flows

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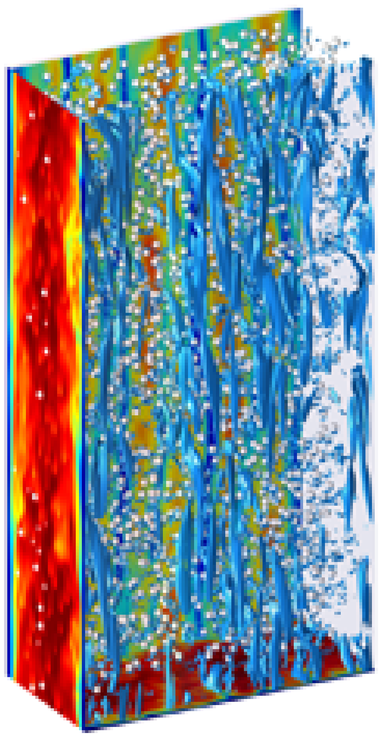
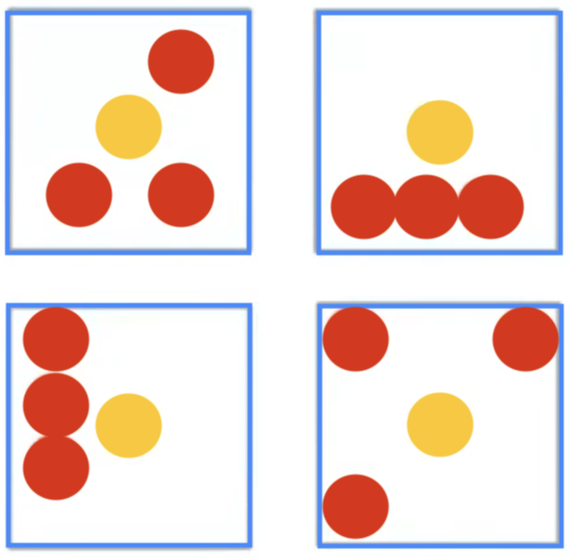
Preamble

- Multiphase flows, such as fluidized beds, are common in various industrial and natural processes.
- Modeling these flows accurately is challenging due to their complex interactions and multiscale nature.
- Mesoscale modeling is commonly used and averages hydrodynamic equations over greater length scales.



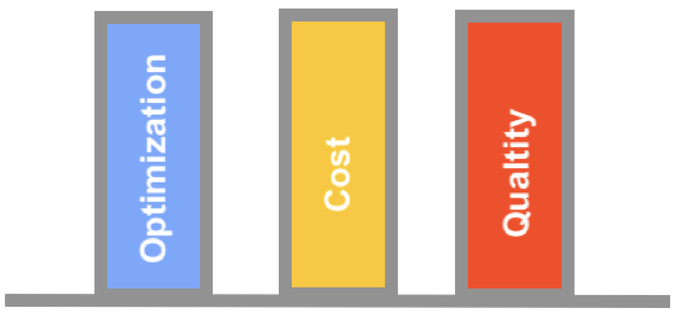
Motivation

Conventional mesoscale modeling neglects microstructure and underestimates the forces in the bed resulting in inaccuracies in predicting bed height and particle velocity fluctuations.

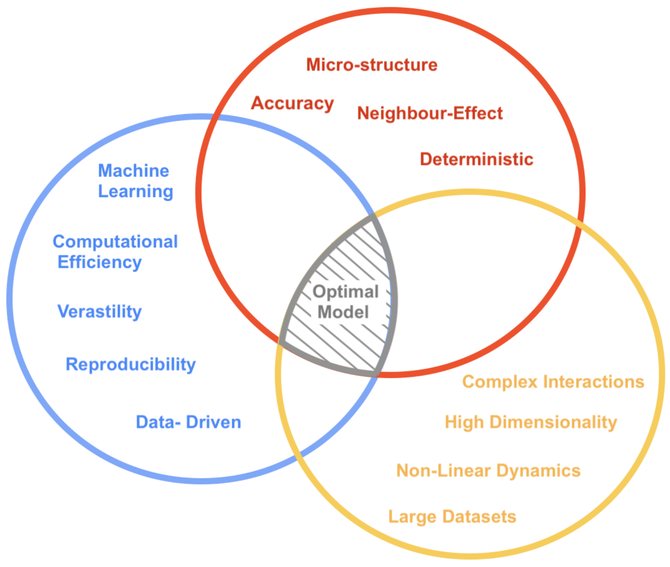


Exploring alternative modeling methods is crucial to:

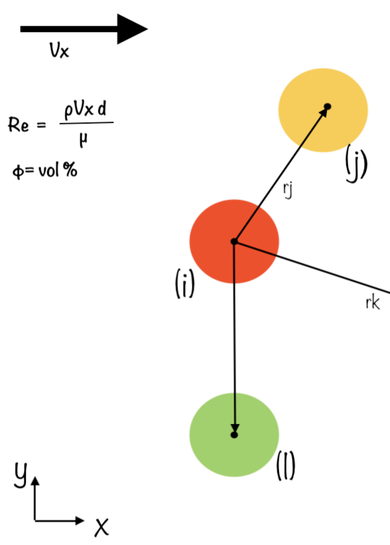
- Optimize Processes: Improve efficiency and productivity.
- Reduce Costs: Decrease energy and material wastage.
- Ensure Product Quality: Maintain consistency and minimize defects.



Objective

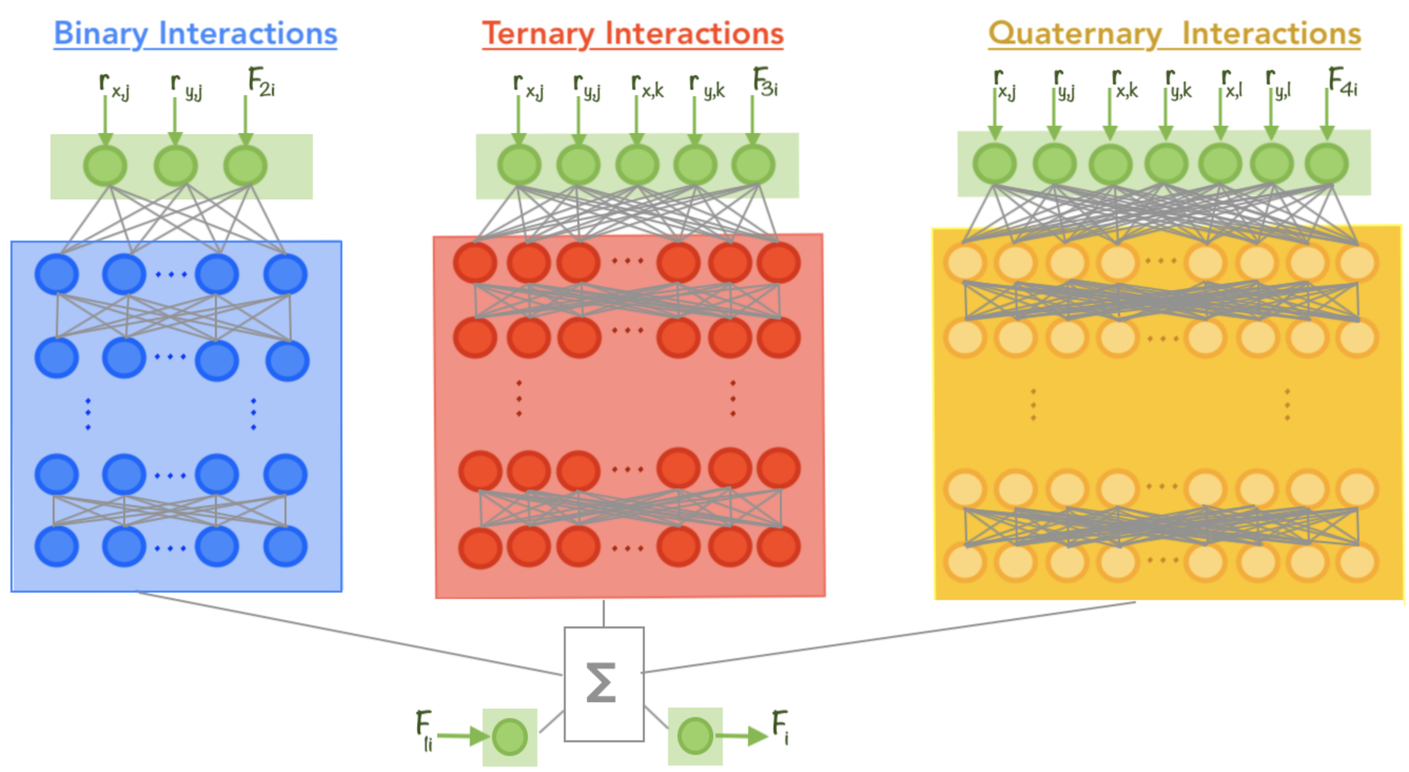


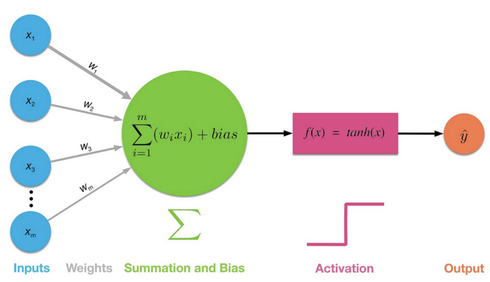
Hierarchical Model



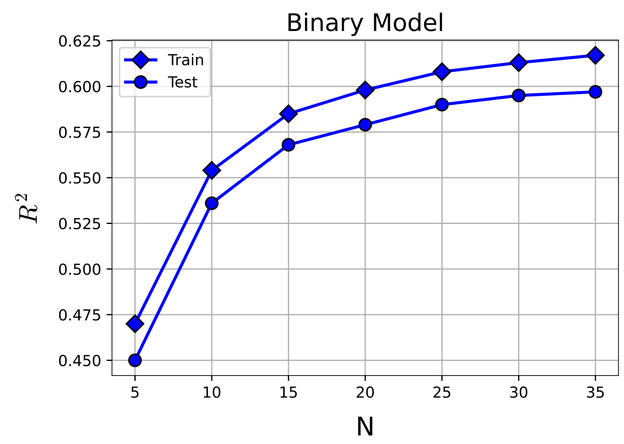
$$\boldsymbol{F}_i(\langle \mathrm{Re}\rangle, \langle\phi\rangle, \{\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_N\}) = \boldsymbol{F}_{1i}(\langle \mathrm{Re}\rangle, \langle\phi\rangle) + \sum_{j=1}^{N} \boldsymbol{F}_{2i}(\langle \mathrm{Re}\rangle, \langle\phi\rangle, \boldsymbol{r}_j) + \sum_{j=1}^{N-1}\sum_{k=j+1}^{N} \boldsymbol{F}_{3i}(\langle \mathrm{Re}\rangle, \langle\phi\rangle, \boldsymbol{r}_j, \boldsymbol{r}_k) + \cdots$$

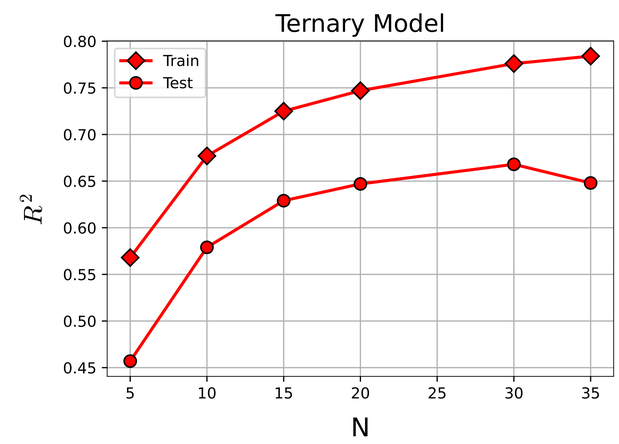
Neural Network design

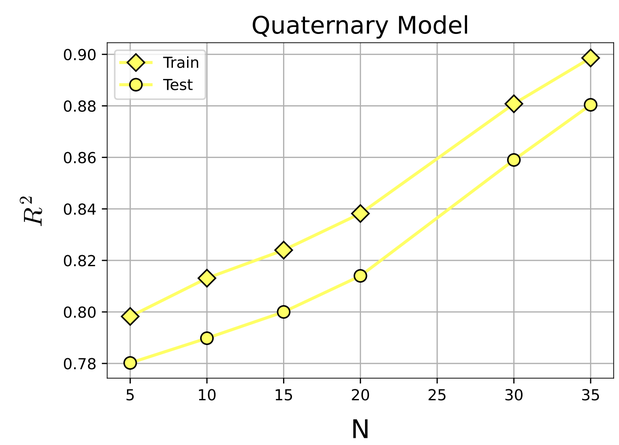




Results







Future Work

- Extension to $3D$ non-homogeneous suspensions.
- Convolutional Neural Network.

Assumptions

- Fully Connected Network
- Unified functionality
- Hyperbolic tangent activation
- Linear Superposition
- Mono-dispersed random distribution

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